

One-factor association of disjoint principal minors of Wishart random matrices

Candidate partial result for arXiv:2311.00202
agent_lane_09, model GPT5.6

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Status. `partial_result_likely_valid; human_review_needed`. The unrestricted conjectures remain open.

1 Source questions

Genest, Ouimet, and Richards [1] consider $\mathfrak{X} \sim \mathcal{W}_p(\alpha, \Sigma)$, with $\alpha > p - 1$, partitioned into diagonal blocks $\mathfrak{X}_{ii}$ of sizes p_i , where $\sum_i p_i = p$. Their Conjecture 1.1 asks whether, for every $\nu_i \geq 0$,

$$\mathbb{E} \prod_{i=1}^d |\mathfrak{X}_{ii}|^{\nu_i} \geq \prod_{i=1}^d \mathbb{E} |\mathfrak{X}_{ii}|^{\nu_i}. \quad (1)$$

The statement is on source page 5:

Gaussian product inequality conjecture for Wishart random matrices

Conjecture 1.1. *Let $p \in \mathbb{N}$ be given. For arbitrary degree-of-freedom $\alpha \in (p - 1, \infty)$ and scale matrix $\Sigma \in \mathcal{S}_{++}^p$, let $\mathfrak{X} \sim \mathcal{W}_p(\alpha, \Sigma)$ be a $p \times p$ Wishart random matrix in the sense of Definition 2.1 below. Write $\mathfrak{X}$ in block matrix form $(\mathfrak{X}_{ij})_{1 \leq i, j \leq d}$, where each block $\mathfrak{X}_{ij}$ has size $p_i \times p_j$, so that $p_1 + \dots + p_d = p$. Then, for every $\nu_1, \dots, \nu_d \in [0, \infty)$, one has*

$$\mathbb{E}(|\mathfrak{X}_{11}|^{\nu_1} \dots |\mathfrak{X}_{dd}|^{\nu_d}) \geq \mathbb{E}(|\mathfrak{X}_{11}|^{\nu_1}) \dots \mathbb{E}(|\mathfrak{X}_{dd}|^{\nu_d}). \quad (1.6)$$

Remark 1.2. Conjecture 1.1 was proved to hold for $d = 2$ by Genest et al. [17], where the joint moment $\mathbb{E}(|\mathfrak{X}_{11}|^{\nu_1} |\mathfrak{X}_{22}|^{\nu_2})$ is shown to be the product of $\mathbb{E}(|\mathfrak{X}_{11}|^{\nu_1}) \mathbb{E}(|\mathfrak{X}_{22}|^{\nu_2})$ and the Gaussian hypergeometric function of matrix argument with certain parameters.

Remark 1.3. One can verify that the expectations in (1.6) are always finite by applying Hölder's inequality to the left-hand side and using the moment formulas for standalone principal minors in Lemma A.2.

Their Conjecture 3.8 asks, for $2 \leq k \leq d$ and $t_i > 0$,

$$\mathbb{P} \left(\bigcap_{i=1}^d \{|\mathfrak{X}_{ii}| \leq t_i\} \right) \geq \mathbb{P} \left(\bigcap_{i < k} \{|\mathfrak{X}_{ii}| \leq t_i\} \right) \mathbb{P} \left(\bigcap_{i \geq k} \{|\mathfrak{X}_{ii}| \leq t_i\} \right). \quad (2)$$

The statement crosses source pages 9–10:

[8] for a survey of its properties.

Remark 3.7. In order to render the upper bound in Corollary 3.4 practicable, explicit expressions for the integrals are derived in Appendix C.

As explained in Remark 4.2 of Edelman et al. [11], an alternative way of showing the lower bound in Corollary 3.4 would be to prove a Gaussian correlation-type inequality for disjoint principal minors of Wishart random matrices. For the aforementioned multivariate Gamma distribution, this Gaussian correlation inequality was proved by Royen in 2014; see Royen [40] and also Royen [41, 42] for some extensions. Consequently, it is natural to conjecture (see Conjecture 3.8 below) that the same inequality holds more broadly for disjoint diagonal blocks of any widths $p_1, \dots, p_d \in \mathbb{N}$.

Gaussian product inequality conjecture for Wishart random matrices

Conjecture 3.8. *In the setting of Theorem 3.3, one has, for every $k \in \{2, \dots, d\}$ and $(t_1, \dots, t_d) \in (0, \infty)^d$,*

$$\mathbb{P} \left(\bigcap_{i=1}^d \{|\mathfrak{X}_{ii}| \leq t_i\} \right) \geq \mathbb{P} \left(\bigcap_{i=1}^{k-1} \{|\mathfrak{X}_{ii}| \leq t_i\} \right) \mathbb{P} \left(\bigcap_{i=k}^d \{|\mathfrak{X}_{ii}| \leq t_i\} \right).$$

The proposition below is a generalization of Theorem 1.1 of Zhou et al. [56] from the multivariate normal setting to the case of disjoint principal minors of Wishart random matrices. It is part of the realm of so-called ‘opposite GPIs’ introduced by Russell and Sun [44], which are described in Section 1. Note that the validity of the proposition is conditional on Conjecture 1.1 being true. Zhou et al. [56] proved the result below unconditionally in the multivariate normal setting, but only when $\nu_2 = \dots = \nu_d = 1$, by

2 Result and proof intuition

Write $D = \text{diag}(D_1, \dots, D_d)$, with $D_i \in \mathbb{S}_{++}^{p_i}$, and write $a = (a_1^\top, \dots, a_d^\top)^\top$. The scale matrices

$$\Sigma = D + aa^\top \tag{3}$$

are precisely coherent one-factor perturbations of block independence: $\Sigma_{ij} = a_i a_j^\top$ for $i \neq j$.

Proof intuition. For block-diagonal Laplace variables, the matrix determinant lemma separates the Wishart transform into independent central-block factors and one scalar factor. The latter is the Laplace transform of a common χ_α^2 variable S . Conditional on $S = s$, the blocks are independent rank-one noncentral Wisharts. A Schur-complement factorization then writes each block determinant as a fixed positive random factor times a noncentral chi-square whose noncentrality is proportional to s . Thus conditional positive moments increase with s , whereas conditional lower tails decrease with s . Products on either side of any split are monotone in the same scalar, and the desired inequalities are Chebyshev covariance inequalities. What remains open is the several-factor case, where the common mixing parameter becomes matrix-valued and this total order disappears.

Theorem 1 (One-factor product association). *Let $p_1, \dots, p_d \geq 1$, $p = \sum_i p_i$, and $\alpha > p - 1$. Suppose Σ has the form (3), and let $\mathfrak{X} \sim \mathcal{W}_p(\alpha, \Sigma)$, partitioned conformally. Put $\Delta_i = |\mathfrak{X}_{ii}|$.*

Let $h_i : (0, \infty) \rightarrow [0, \infty)$ be Borel functions whose required expectations are finite. If all h_i are nondecreasing, or all are nonincreasing, then for every split $A \sqcup B = \{1, \dots, d\}$,

$$\mathbb{E} \prod_{i=1}^d h_i(\Delta_i) \geq \mathbb{E} \prod_{i \in A} h_i(\Delta_i) \mathbb{E} \prod_{i \in B} h_i(\Delta_i). \quad (4)$$

Consequently (1) holds for all real $\nu_i \geq 0$, and (2) holds for all $t_i > 0$.

3 Proof

We use the Wishart convention of [1, 4]:

$$\mathbb{E} e^{-\text{tr}(T\mathfrak{X})} = |I + 2T\Sigma|^{-\alpha/2}, \quad T \geq 0. \quad (5)$$

3.1 An exact common-scalar mixture

Take $T = \text{diag}(T_1, \dots, T_d)$, $T_i \geq 0$, and set

$$q_i(T_i) = a_i^\top T_i (I + 2D_i T_i)^{-1} a_i \geq 0.$$

The matrix determinant lemma and the push-through identity give

$$|I + 2T(D + aa^\top)| = \prod_{i=1}^d |I + 2T_i D_i| \left(1 + 2 \sum_{i=1}^d q_i(T_i) \right). \quad (6)$$

Let $S \sim \chi_\alpha^2$. Conditional on $S = s$, let $\mathfrak{Y}_1, \dots, \mathfrak{Y}_d$ be independent noncentral Wishart matrices whose transforms are

$$\mathbb{E} \left[e^{-\text{tr}(T_i \mathfrak{Y}_i)} \mid S = s \right] = |I + 2T_i D_i|^{-\alpha/2} e^{-sq_i(T_i)}. \quad (7)$$

These laws exist because $\alpha > p - 1 \geq p_i - 1$; after whitening their noncentrality matrices are $sD_i^{-1/2} a_i a_i^\top D_i^{-1/2}$, of rank at most one. Since

$$\mathbb{E} e^{-uS} = (1 + 2u)^{-\alpha/2},$$

averaging the product of (7) and using (6) yields (5) for every block-diagonal T . Uniqueness of Laplace transforms on products of positive-definite cones gives

$$(\mathfrak{X}_{11}, \dots, \mathfrak{X}_{dd}) \stackrel{d}{=} (\mathfrak{Y}_1, \dots, \mathfrak{Y}_d). \quad (8)$$

This is valid for every real $\alpha > p - 1$, not only integer degrees of freedom.

3.2 Rank-one determinant monotonicity

Lemma 1. Let $r \geq 1$, $\beta > r - 1$, and let Y_λ be a noncentral Wishart matrix with degrees of freedom β , scale I_r , and noncentrality $\lambda e_1 e_1^\top$. There are independent random variables

$$Q_\lambda \sim \chi_\beta^2(\lambda), \quad C \sim \mathcal{W}_{r-1}(\beta - 1, I_{r-1}),$$

with $|C| = 1$ when $r = 1$, such that

$$|Y_\lambda| \stackrel{d}{=} Q_\lambda |C|. \quad (9)$$

In particular $|Y_\lambda|$ is stochastically nondecreasing in λ .

Proof. The rank-one noncentral Wishart density is the central density multiplied by

$$e^{-\lambda/2} {}_0F_1\left(\cdot; \frac{\beta}{2}; \frac{\lambda y_{11}}{4}\right).$$

Indeed the matrix hypergeometric series reduces to the scalar series because the argument has rank one; this is the standard noncentral Wishart density formula [4].

Under the central law, write

$$Y = \begin{pmatrix} x & z^\top \\ z & C + zz^\top/x \end{pmatrix}.$$

Then $|Y| = x|C|$. The change of variables $z = \sqrt{x}u$ factors the central density into the densities of

$$x \sim \chi_\beta^2, \quad u \sim N_{r-1}(0, I), \quad C \sim \mathcal{W}_{r-1}(\beta - 1, I),$$

independently. The noncentral density multiplier depends only on x and is exactly the likelihood ratio of $\chi_\beta'^2(\lambda)$ relative to χ_β^2 . It therefore replaces x by Q_λ without changing the independent law of C , proving (9).

Finally, if $N_\lambda \sim \text{Poisson}(\lambda/2)$, then

$$Q_\lambda \mid N_\lambda \sim \chi_{\beta+2N_\lambda}^2.$$

For $\lambda' \geq \lambda$, couple $N_{\lambda'} = N_\lambda + \tilde{N}$ with an independent $\tilde{N} \sim \text{Poisson}((\lambda' - \lambda)/2)$, and add independent chi-squares of two degrees of freedom. This couples $Q_{\lambda'} \geq Q_\lambda$, hence also $Q_{\lambda'}|C| \geq Q_\lambda|C|$. $\square$

For block i , whiten by $D_i^{-1/2}$ and rotate $D_i^{-1/2}a_i$ to the first coordinate. Lemma 1 and (7) show that, conditional on $S = s$,

$$|\mathfrak{Y}_i| \stackrel{d}{=} |D_i| Q_{i,s} B_i, \quad Q_{i,s} \sim \chi_\alpha'^2\left(s a_i^\top D_i^{-1} a_i\right), \quad (10)$$

where $B_i > 0$ has a law independent of s and is independent of $Q_{i,s}$. Hence the conditional law of $|\mathfrak{Y}_i|$ is stochastically nondecreasing in s .

3.3 Scalar association

Set

$$m_i(s) = \mathbb{E}[h_i(|\mathfrak{Y}_i|) \mid S = s].$$

By (10), all m_i are nondecreasing when all h_i are nondecreasing, and all are nonincreasing when all h_i are nonincreasing. Conditional independence and (8) give

$$\mathbb{E} \prod_{i=1}^d h_i(\Delta_i) = \mathbb{E} \prod_{i=1}^d m_i(S).$$

Put $M_A = \prod_{i \in A} m_i$ and $M_B = \prod_{i \in B} m_i$. These two functions have the same monotonicity direction. If S' is an independent copy of S , Chebyshev's covariance identity gives

$$\text{Cov}(M_A(S), M_B(S)) = \frac{1}{2} \mathbb{E}[(M_A(S) - M_A(S'))(M_B(S) - M_B(S'))] \geq 0.$$

This is precisely (4).

Taking $h_i(x) = x^{\nu_i}$ and iterating splits proves (1). Taking $h_i(x) = \mathbf{1}_{\{x \leq t_i\}}$ and the split $A = \{1, \dots, k-1\}$, $B = \{k, \dots, d\}$ proves (2). $\square$

4 Verification and limitations

- The proof was checked separately at the three structural points: the determinant-lemma transform, the rank-one density tilt, and the scalar covariance step.
- The accompanying program compares direct SciPy Wishart samples with the determinant mixture for randomized noninteger degrees of freedom, and tests both inequalities. This is a smoke test, not part of the proof.
- The result includes arbitrary block sizes, arbitrary d , every real $\alpha > p - 1$, every real $\nu_i \geq 0$, and every $t_i > 0$.
- It does not settle a general Σ . With several latent factors, the conditional noncentrality is governed by a matrix Gram parameter; varying one latent factor can change cross terms in both directions, so the scalar stochastic-order argument does not iterate.

Novelty check. The four local run indexes had no hit for arXiv:2311.00202 or this route. Bounded searches covered the exact conjecture labels, source title, *one-factor Wishart Gaussian product inequality*, *rank-one noncentral Wishart determinant stochastic monotonicity*, and papers citing the source. The two-block theorem of [2] and the July 2026 finite Stein recursion of [3] were inspected. No statement of Theorem 1 was found. Novelty is therefore plausible, not certified.

Human-review recommendation. Verify especially the noncentral-Wishart parameter convention in (7), the rank-one density reduction in Lemma 1, and Laplace-transform uniqueness for the joint diagonal-block law. If those pass, retain this as a substantial partial answer to both source conjectures.

References

- [1] C. Genest, F. Ouimet, and D. Richards, *On the Gaussian product inequality conjecture for disjoint principal minors of Wishart random matrices*, Electron. J. Probab. 29 (2024), arXiv:2311.00202v3, doi:10.1214/24-EJP1222.
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- [3] R. E. Gaunt and F. Ouimet, *On Wilks’ problem: Exact recursive formulas via Stein’s method for the joint moments of disjoint principal minors of Wishart random matrices*, arXiv:2607.03984v1 (2026).
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